

Graduate Handbook: Ph.D. in Applied Biology

Department of Biological Sciences
University of Massachusetts Lowell

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Department Of Biological Sciences

The Department of Biological Sciences prepares students for a variety of careers that contribute to society's understanding and advancement of health, education, and environmental issues. According to the United States Department of Labor, workers with a graduate STEM degree earn a higher median weekly salary and annual salary than those with a bachelor's degree. Students who earn their graduate degrees from the Department of Biology at UML can expect to learn skills and techniques that are broadly applicable to a wide variety of fields in the disciplines of biomedicine and environmental research. Many graduates will become employed in academic and industrial research institutions, while others will enhance biomedical research or environmental field studies. The Department of Biological Sciences is committed to educating its students about diverse topics pertaining to biology and their practical and ethical impact on the world we live in.

Ph.D. in Applied Biology: The program integrates foundational areas of biology with focused interdisciplinary biological research and preparation for career advancement. It combines fundamental and rigorous training in the pursuit of new knowledge regarding the mechanisms of biological processes coupled with the skills needed to apply this knowledge in the workforce. In addition to traditional academic careers in research and education at colleges and universities and academic research hospitals and institutes, our graduates are also prepared for non-academic careers.

For more information, please check the website:
https://www.uml.edu/sciences/biology/programs-of-study/phd-applied-biology.aspx

Any questions can be directed to our Graduate Coordinators:

Dr. Jennifer Fish
Associate Professor
Graduate Coordinator
Jennifer_Fish@uml.edu
978-934-5337
Dr. Fish assists students with course selections, answers questions about degree requirements, and ensures students fulfill their academic responsibilities.

Dr. Natalie Steinel
Associate Professor
Graduate Coordinator for Admissions
bio_grad_admissions@uml.edu
978-934-2833
Dr. Steinel assists students with graduate applications and reviews all applicants to the M.S. and Ph.D. programs. Questions about applications are directed to her.

Statement of diversity, equity, and inclusion

We welcome all who wish to study the Biological Sciences at the University of Massachusetts Lowell and support their growth as students and researchers. Our graduate program values diversity and seeks talented students from diverse backgrounds. We do not discriminate on the basis of age, race, culture, disability status, gender identity or expression, sexual orientation, religion, class, financial aid, national origin, native language, work experience, and/or veteran status. We also recognize the intersectionality of these groups and that individuals do not fall exclusively into a single category. We expect all of our students to demonstrate the same commitment to diversity, equity, and inclusion, and we do not tolerate discrimination of any kind.

UML administrative staff

Biology Department
For answers about GA/RA/TA contracts, and to schedule an appointment with the Department Chair, please contact: Irma Silva - Department Administrator
Office Hours: Monday through Friday: 7:30 a.m. - 4:00 p.m.
Email: Irma_Silva@uml.edu
Olsen Hall 234
North Campus
Phone: 978-934-2661

Graduate Admissions Office
For help with general graduate admission issues, contact:
Phone: 978-934-2390
Toll Free: 800-656-GRAD
Fax: 978-934-4058
Email:Graduate_Admissions@uml.edu

Registrar's Office
For help with course registration, contact the solution center:
Email: TheSolutionCenter@uml.edu
Phone: 978-934-2000
University Crossing
220 Pawtucket St., Suite 131
Lowell, MA 01854-5141

International Student and Scholars Office
For help with visas and other questions pertaining to international student affairs, contact:
Email: ISSO@uml.edu
Phone: 978-934-2383
Office Hours: Monday through Friday: 9 a.m. - 4:30 p.m.
Cumnock Hall, Suite 208
31 University Avenue
Lowell, MA 01854

Description of Ph.D. in Applied Biology program

The program combines fundamental and rigorous training in the pursuit of new knowledge regarding the mechanisms of biological processes coupled with the skills needed to apply this knowledge in the workforce. In addition to traditional academic careers in research and education at colleges and universities and academic research hospitals and institutes, our graduates are also prepared for non-academic careers. To provide students an area of focus within the broad field of biology, the Applied Biology program includes four options:

- Biomedical Science
- Developmental & Evolutionary Biology
- Quantitative Biology & Biophysics
- Cellular & Molecular Biology

Students select their option anytime during their first two years in the program.

Academic expectations of graduate students

You are in a unique position as a Ph.D. student at UML. Undergraduate students look up to you for both knowledge and advice, and faculty members have higher expectations of you than undergraduate students. Therefore, graduate students are held to a rigorous standard of professional and ethical conduct during their academic program. We expect graduate students to maintain integrity with their research and in the lab. Below, we outline specific academic expectations for our graduate students.

1. Take your coursework seriously. Show up for class prepared and on time. You are a role model for undergraduate students, and arriving unprepared and/or late for any class is unprofessional and unacceptable. You must notify the instructor of any absences ahead of time. Missing any class at the beginning of a semester is unacceptable. In the event that you miss the first class meeting, you are not guaranteed enrollment in the class, and you may be dropped from it at the professor's discretion.

2. Earn a grade of B (3.0) or better. You are not allowed to count more than nine credits below B towards your Ph.D. degree (Please note that a B- is below B). Additionally, no graduate degree will be awarded to a student whose cumulative average for course work is below 3.0. Remember, as a graduate student, you are held to high expectations by the faculty. Grades possible are: A+(4.000), A(4.000), A-(3.700), B+(3.300), B(3.000), B-(2.700), C+(2.300), C(2.000), F(0.000)

3. Demonstrate professional behavior in all of your interactions with other students (graduate and undergraduate), faculty, and staff.

4. Complete your program in a timely manner. Ph.D. students are expected to fulfill the requirements of their program in a maximum of 6 years. Longer timelines for extenuating circumstances will require approval by your advisor and the graduate coordinator. Extended absences (>1 week) should be scheduled ahead of time and brought to the attention of your faculty advisor, graduate coordinator, and course professor immediately. Graduate students should inform the appropriate faculty of the need for the absence and the expected date of return. Meet with your advisor on a regular basis to ensure that you are enrolled in the appropriate courses and following the recommended guidelines for your academic program.

Policies for Teaching Assistants and Research Assistants

All graduate students working for the University of Massachusetts Lowell are designated as employees and represented by the **Graduate Employee Organization Local 1596 UAW**. Students are encouraged to examine their union contract and speak with union representatives if they have questions about their employment. They are also encouraged to speak with the Graduate Coordinator, their Dissertation Advisor, or the Department Chair of Biological Sciences if they have specific questions about their roles in the department.

1. Assistantships may be provided to students in good academic standing during their time in the Department of Biological Sciences. The student must be full time and registered for exactly nine credits. Students or the mentor are responsible for payment for any additional credits. In your final semester, enrollment in fewer than nine credits is allowed with submission of a reduced course load form. https://www.uml.edu/graduate-student-services/ta-ra/ta-ra-forms.aspx.

2. All students must remain in good academic standing (earn minimum 3.0 GPA each semester) to be awarded a TA/RA during their time at UML. Per the Graduate Student Collective Bargaining Agreement, students who have earned an F, U, or have an INC on their transcript are not eligible for TAs. See UML Graduate Academic Standing policies here: https://www.uml.edu/catalog/graduate/policies/academic-standing.aspx.
See Graduate Union CBA agreements here: https://www.uml.edu/hr/labor-relations/ta-labor.aspx.

3. Students on a TA must register for coursework (rather than research credits) until their coursework is completed. Exceptions may be made on a case-by-case basis for individuals that need one or two dissertation credits to reach the required nine credits per semester. Students on RAs are also strongly encouraged to complete their coursework in the first two years of their dissertation to encourage timely progression to candidacy.

4. The level of assistantship received is commensurate with experience (as outlined in the union contract). Level 1 stipends are awarded to first year doctoral students. Level 2 stipends are awarded to: matriculated doctoral students who have demonstrated satisfactory degree progress after their first year and have been recommended by department/college reviewing authorities; new doctoral students holding a master's degree; and new doctoral students with special achievements upon recommendation by the college dean. Level 3 stipends are awarded to graduate students who have reached doctoral candidacy (i.e. passed both the qualifying exam and proposal defense) and are enrolled in dissertation research.

5. Per union contract, students must work 18 hours per week for a full time assistantship.

6. TAs will be assigned as available following these priorities: 1) Satisfactory progress, evaluated first by grades, then by research progress; 2) Ph.D. students; 3) MS students enrolled in thesis; 4) MS students enrolled in project credits; 5) Other factors. In addition, the following factors will be considered positively: faculty advisor is pre-tenure, faculty advisor is tenure-track, faculty advisor can no longer serve, special circumstances approved by the Chair and Graduate Coordinator. The following factors will be considered negatively: faculty advisor is not in Biological Sciences, faculty advisor has funds to support the student on an RA, time in program (past 6 years for Ph.D., past 2.5 years for MS), has held TA-ship for two years, research group already has 2 TAs.

Description of the roles and responsibilities of TAs and RAs

Teaching Assistantships: A Teaching Assistant (TA) is a student who is enrolled full time and employed by the department for the purposes of assisting professors in lecture and laboratory courses. A TA's responsibilities vary based on the course, and should be discussed with the instructor of record at the beginning of the semester. You must speak to your supervisor ahead of time if you expect an interruption in your TA responsibilities.

Responsibilities may include the following:

1. Assist or help coordinate activities such as exams, quizzes, and problem sessions.
2. Tutor students enrolled in the course, and making yourself available in person or via email.
3. Lead instructional activities in laboratory courses including the setup, coordination, and dismantling of experiments and other laboratory activities such as dissections.
    a. Note: TAs must never leave their classroom unattended while a class is in session except for brief bathroom breaks. In addition, classes cannot be canceled without your supervisor's approval. For extended absences while a course is in session, the TA must obtain permission and assistance.
4. Assist with or lead grading and proctoring activities.
    a. Note: grading must not be delayed; this includes individual assignments as well as posting final grades.
5. Attend weekly preparation sessions.

Research Assistantships : Research assistantships (RA) are only available through special arrangements with individual faculty research advisors. Granting of such assistantships is at the discretion of individual faculty members, each of whom may have their own specific requirements that students must fulfill. Working in a research laboratory is not a guarantee of being granted a research assistantship.

An RA's responsibilities may include the following:

1. Gathering and analysis of data for the benefit of the student's or faculty's research.
2. Development of models for theoretical analysis.
3. Production of research publications.
4. Production of research reports (e.g., for grants or presentation at conferences).
5. Attendance at laboratory meetings.
6. Attendance at journal clubs.
7. Attendance at departmental seminars.

Procedures for changing programs

Students accepted into a Master's program who decide to remain for a doctorate degree (Ph.D) but want to first complete their master's degree.

1. The student must complete all required courses, compile a 3.0 grade point average, and successfully defend his/her thesis, if required.
2. The student must complete the Graduate School clearance process for the master's degree.
3. A student is prohibited from enrolling in doctoral research until he or she has completed the clearance process for the M.S. degree.
4. The student must then apply to the doctoral program by completing the standard application process.

Official admission into a doctoral program and receipt of a letter of acceptance are contingent upon completion of the clearance process for the M.S. degree.

Students accepted into a doctoral program who elect to obtain the Master's degree and leave the university:

1. The student must file an Academic Petition requesting to be changed from the doctorate to the master's degree program.
2. The student's research advisor must submit to the Records Office, grade change forms withdrawing the student from doctoral dissertation and adding master's thesis for the appropriate number of credits.
3. The student must complete all required courses for the master's degree, earn a minimum 3.0 grade point average, successfully defend their thesis, and complete the clearance process at the Graduate School.
4. All graduate courses (and undergraduate coursework used for graduate credit), whether taken for the original doctoral program or for the master's degree, will be included in the grade point average and listed under the master's degree program.

Selection of the Dissertation Advisor (PI)

The Applied Biology Ph.D. program is a direct admit program, meaning that prospective students must talk to potential advisors before application to the program and submit a letter of sponsorship from the potential advisor with their application. When admitted, students are admitted to a specific research advisor's lab. Most of the Applied Biology Ph.D. advisors are in the Department of Biological Sciences. Supervision by advisors outside of the Biological Science department will be considered on an individual basis by the Graduate Policy and Education Committee and the Graduate Coordinator.

To accept Applied Biology Ph.D. students, advisors must agree to the following terms:

1. PIs recruiting Applied Biology Ph.D. students must commit to financially supporting them until they are eligible for a course-based MS degree. This commitment will be included in the student's acceptance letter. Support includes providing a summer salary and either a TA or RA until the student is eligible for an MS degree. However, should the student opt to graduate with a MS degree, it should be noted that TAs are not guaranteed for MS students. Therefore, if no TA is available, then the PI will support the student with an RA until they are eligible to leave with a MS degree (provided that the student makes acceptable progress with coursework).
2. PIs who do not support students as stated above may face restrictions on future TA assignments and/or admittance of students into the Applied Biology Ph.D. program.
3. For extenuating circumstances where a student fails to meet minimum expectations (e.g., attend scheduled meetings, perform TA duties, maintain acceptable progress towards their degree), PIs can dismiss a student by:
    a. Filling out semester TA/RA evaluations to reflect performance issues.
    b. Documenting the issues/concerns with Performance Improvement Plans (PIP) approved by the Graduate Coordinator (see Appendix V). This plan should set clear expectations, outline specific and measurable objectives/deliverables, and establish a timeline for check-ins to discuss progress.
    c. Providing a reasonable time frame (generally 1 semester) for the student to address issues/concerns raised in the PIP and bring performance up to the expected level.

How to change research advisors: The student must submit an academic petition with all required signatures of the Graduate Coordinator and new dissertation advisor. The petition should clearly state that the previous thesis or dissertation project work is satisfactory and will carry over. Theses and dissertation advisors are recorded in SIS. An academic petition is required to change advisors to ensure that records are accurate. The Graduate Coordinator determines whether previous work will or will not be counted towards the degree and notes this on the petition.

Curriculum description

The curriculum for the Ph.D. in Applied Biology consists of a combination of required courses, core competency courses, advanced electives and thesis research credits, leading to the oral defense of a written thesis based on independent research. At the start, all students must take the series of core courses included in the Applied Biology Program to ensure that all students develop key skills needed to be successful in completing their independent research. Upon entry into the program, each student's competency in foundational areas of biology will be evaluated and courses will be assigned based on any identified deficiencies.

Major Required (Core) Courses (Total courses required = 7)		
Course Number	Course Title	Credit Hours
BIOL.6040	Professional Communication in Science & Technology	3
BIOL.6060	Applied Biology I	1
BIOL.6070	Applied Biology II	1
BIOL.6030	Graduate Colloquium Biology	1
BIOL.6030	Graduate Colloquium Biology	1
BIOL.6050	Graduate Proposal Writing Seminar	1
BIOL.7210	Special Problems in Biology*	3
	SubTotal # Core Credits Required	11
Other/Elective Course Choices (Total courses required = 7) (*see below lists)*		
BIOL.XXXX	Competency Course**	3
BIOL/other	Competency Course or Advanced Elective***	3-4
BIOL/other	Special Problems in Biology or Advanced Elective	3
BIOL/other	Advanced Elective	3-4
BIOL/other	Advanced Elective	3-4
BIOL/other	Advanced Elective	3-4
BIOL/other	Advanced Elective	3-4
	SubTotal # Elective Credits Required	21-26
Research credits (to make 60 total credits)		
BIOL.7530	Ph.D. Dissertation Applied Biology	23-28
Curriculum Summary		
	Total number of courses required :	14
	Minimum number of course credits required:	32 (max 37)
	Minimum number of research credits required:	23 (max 28)

*** Special Problems in Biology** can include either 1. doing rotations with research faculty other than the dissertation advisor, or 2. taking minicourses with faculty other than the dissertation advisor (or, with permission from Graduate Coordinator, with the student's dissertation advisor). Three credits total are necessary to fulfill this requirement. When seeking rotations, students should ask faculty who have complementary interests, could teach them a useful skill, and/or broaden the intellectual scope of their project.

**** Core Competency Courses** will be defined by each dissertation advisor in collaboration with the PhD student. In most cases, students will take up to two core competency classes.

***** Advanced Electives** can be selected from the graduate course catalog or through discussion with Graduate Coordinator. Students are required to complete at least four advanced courses in their option and two courses outside of their option. Electives outside of the department can be used if approved by the graduate coordinator. A maximum of 9 credits may be completed at other institutions via the UMass Inter-Campus Course Exchange Program with prior permission from the Graduate Coordinator. ***Optional Industry Internship*** – Students may participate in an industry internship for course credit (BIOL.7070 "Internship Biology") at a biotech/biomedical company that complements their option specialization. Prior to initiating an internship, students must have completed at least one traditional academic year (fall/spring or spring/fall), have completed 18 credits and be in good academic and judicial standing. Students on F-1 or J-1 nonimmigrant status must also obtain prior approval from the International Students & Scholars Office.

All international students on F-1 or J-1 visas must register as full-time students (at least 9 credits). Any variance from this policy must be approved by submitting the appropriate forms (https://www.uml.edu/isso/reduced-course-load/how-to-apply.aspx). Students in the final semester can petition for a reduced course load by submitting a reduced course load form.

Suggested timeline

Year 1

- In the first year of the program, students should take 9 credits per semester. Students should prioritize taking competency and core courses, especially Applied Bio I and II.
- Under the guidance of their Ph.D. supervisor, students develop an individual dissertation research project plan.

Year 2

- In year 2, students should complete their remaining coursework, taking 9 credits per semester.
- Formation of the Dissertation Committee: In the fall of the second year, the advisor and student will work together to select a committee (in consultation with GPEC as needed) composed of 3-4 faculty members, who will oversee the student's progress.
- Comprehensive Qualifying Examination: In the late spring of the second year, the student must take an oral qualifying examination. Advanced students in good standing may petition the grad coordinator to take this exam at the end of year 1.

Year 3

- In year 3, students concentrate on their research and prepare their written dissertation proposal.
- The written proposal must be defended in an oral presentation before the start of their 4th year. The defense must be advertised two weeks beforehand.

Years 4 and beyond

- In years 4 and beyond, students will work closely with their advisor and committee to complete their research and dissertation.
- After successfully defending their proposal, each student will meet with their Dissertation Advisory Committee a minimum of once every 12 months to provide updates on progress as well as immediate and long-term plans.
- Once students have completed a significant body of publishable research, they will prepare a written dissertation and present a public oral defense to their Dissertation Advisory Committee (see section on dissertation requirements).

Note: students who have completed all requirements except the writing and defense of the dissertation and do not need university resources can register for continued matriculation (CONT.7010) and pay a fee until they graduate.

Summary of graduation requirements

1. Completion of 60 credits: 30 cr of coursework and 30 cr of dissertation research.
2. Pass the Qualifying Exam.
3. Provide annual updates to a Dissertation Advisory Committee.
4. Write a Dissertation Proposal and pass the oral defense of the proposal.
5. Deliver a written dissertation and pass the oral defense of the dissertation.
6. Submit the dissertation.
7. Have a minimum cumulative GPA of 3.0.

Dissertation committee

Composition: The dissertation committee consists of 3-4 faculty members (not including the advisor), with at least two from the Biological Sciences Department at the University of Massachusetts Lowell. An outside expert from industry or another university may be a member of the committee, but that individual must possess academic credentials which would qualify them to serve as a member of the University of Massachusetts Lowell faculty. The student should reach out to prospective committee members at least four months before their qualifying exam. A dissertation enrollment form (see appendix 1) should be filled out by the student as they form their committee and turned into the Grad coordinator.

Responsibilities: The responsibilities of the Dissertation Committee shall be to:

1. Administer and evaluate the Qualifying Exam
2. Evaluate the written proposal and oral proposal defense
3. Supervise the progress of the dissertation through committee meetings
4. Evaluate the written dissertation and oral defense of the dissertation
5. Report the completion of all dissertation requirements to the department and Grad School

Qualifying examination

Goals: The qualifying examination will evaluate the students' critical thinking and the ability to apply scientific principles to address research problems.

Applied Biology Ph.D. Qualifying Exam Learning Objectives:

The qualifying examination will evaluate the students' critical thinking and the ability to apply scientific principles to address research problems, as detailed in the following learning objectives (a rubric will be provided to the student 2 weeks prior to the exam):

1) Core knowledge: Demonstrate familiarity with core knowledge for the student's discipline (as determined by the student's qualifying exam committee and built around/represented in the paper selection)

2) Scientific literacy: Demonstrate knowledge of literature relevant to the chosen articles, and the application of scientific methods to test hypotheses and understand biological processes

3) Critical thinking:

- Application of scientific literacy, including the ability to synthesize and integrate fundamental biological concepts presented in their coursework or literature
- Demonstrate proficiency in self-directed learning and ability to extrapolate based on their current knowledge (ie: How would one go about finding missing information? How would one formulate a hypothesis and test it?)

4) Experimental Design and Data Analysis:

- Demonstrate understanding of the significance, approach, and analysis for chosen papers
- Significance: demonstrate understanding of the context of the work and implications of the authors' hypothesis
- Approach: describe experimental strategies and explain why specific methods were chosen (rather than possible alternatives); demonstrate understanding of appropriate controls (or lack of)

5) Analysis: demonstrate ability to interpret data presented and draw conclusions: Clearly explain any statistical analyses applied to demonstrate validity of conclusions. Discuss appropriateness of figures/tables and possible alternatives to communicate the findings.

6) Oral Communication: The student can organize and clearly explain their thoughts and provide rationale/support for their ideas.

Planning and scheduling of the qualifying exam:

1) Please set up a QE committee.

- Your QE committee must be comprised of three members. Typically, members of your QE committee are also part of your dissertation committee.
- Your PI cannot be part of the QE committee. So, if your dissertation committee only consists of your PI and two others, you should identify another faculty member who can serve as the third member of your QE committee.
- At least two members of your committee must be faculty in the Biological Sciences department.

2) The committee must choose a committee chair. The chair will be responsible for enforcing the completion of the paperwork related to the exam, moderating the exam, and generally keeping things on track. (The GPEC rep's role is to oversee this process.)

3) Please share with the GPEC rep the names of individuals who are on your committee and specify the chair. This will help the GPEC rep communicate with them the details they need to know regarding this exam.

4) Please set a date, time for your QE, and also reserve a conference room (note that external committee members may attend over Zoom if necessary). When scheduling the exam, please ensure that:

- All QEs must be scheduled by the deadline stated by the GPEC rep in the informational email.
- All QEs must occur within the timeframe stated by the GPEC rep. If an exception is needed due to fieldwork or travel, this must be approved by the GPEC rep well in advance of the scheduling deadline.
- The date of the QE must be compatible with all three committee members and the GPEC rep (use a poll to determine an appropriate time).
- The time period booked for the exam should be at least 3 hours long.
- If external committee members need to attend over Zoom, please schedule appropriately.
- Your committee members, PI, and the GPEC rep are notified of any major changes.

5) Please make sure that you, your PI, and your QE committee members have read the instructions for the QE (see below). Please send these instructions and the QE rubric to your committee members. They will receive these again as a reminder as well from me once you have updated me with their information.

Structure of the exam

1) Your PI will coordinate with your QE committee members to select three primary research papers (one per committee member). Your PI will send you these papers exactly two weeks before your scheduled QE (please be sure that your PI cc's me as the GPEC rep on that email). These three papers will serve as the 'study material' you are expected to master for your oral exam.

2) During the exam, you will be asked questions aimed at gauging your understanding of the paper you were asked to read/study (see below for more details: Applied Biology Ph.D. Qualifying Exam Guidelines).

3) The logistics of how the exam is conducted are flexible. But generally:

- All three papers must be discussed during the exam.
- All members should have adequate time to evaluate the student by asking questions.
- The initial discussion of science/papers should be at least 90 minutes long (we recommend at least 30 minutes per paper, but longer discussion, if desired by members of the QE committee, is fine).
- If necessary, the committee can choose to ask general questions after each of the three papers has been discussed.
- The QE Chair should ensure that there is enough time left at the end of the QE for private committee discussion (with the student sent out of room), as well as time to summarize the committee's evaluation with the student at the end of the exam.
- If the student requests, PIs can sit in on the QE as a silent witness – they cannot participate in the questioning, and must not communicate with the student during the QE. If the student elects to have their PI participate, they should notify the GPEC rep before the exam.

Hang in there! We are almost at the end. Just one more section to go!

Outcome: There are three possible outcomes of the exam: Pass, Conditional Pass, and Fail. The QE Chair will deliver the results to the student by collecting statements/suggestions from committee members. The chair will also initiate and sign the **Report of Examination Form**, available via Docusign (see Appendix III). The report must then be signed by the PI/advisor, the committee members, the student, and the Graduate Coordinator after completion of the qualifying exam. The committee will set the terms and timeline for the student to convert a Conditional Pass to a Pass. Students who fail the qualifying exam will be given one opportunity to retake the exam in the early fall of the next year. Students who fail the exam a second time will be unable to advance to candidacy for the doctoral degree. If appropriate, they may be directed to complete an M.S. degree.

Please let the GPEC rep know if you have any further questions about this process.

Written dissertation proposal

The dissertation proposal should be written either in the format of an NSF or NIH grant proposal, with input from the research advisor. Below are summaries of these guidelines. The proposal will outline the student's project and provide all the necessary background and justification.

Formatting:
1. Standard 8.5" x 11" page size
2. 12-point, Times New Roman font
3. 10-point font may be used for references, footnotes, figure captions, and text within figures
4. 1" margins on all sides
5. Single-spaced (approximately 5 lines per inch) or greater line spacing. Do not use line spacing options such as "exactly 12 point," that are less than single spaced.

1. Cover page with proposal title, student name, and advisor name

2. Project Summary: A single page overview of the research.

- NIH Specific Aims page, which summarizes the context and significance of the work, a short description of aims, and a summary of impact. Details and examples can be found:
  - https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6133727/
  - https://www.biosciencewriters.com/NIH-Grant-Applications-The-Anatomy-of-a-Specific-Aims-Page.aspx
- NSF Project Summary page often consists of three parts: one paragraph that describes the research; one paragraph on the intellectual merit of the proposed research; and one paragraph on the broader impacts of the proposed research. **Intellectual Merit:** The Intellectual Merit criterion encompasses the potential to advance knowledge; and **Broader Impacts:** The Broader Impacts criterion encompasses the potential to benefit society and contribute to the achievement of specific, desired societal outcomes." Details and examples can be found:
  - https://grantwriting.stanford.edu/project-summary/
  - https://granttrainingcenter.com/blog/important-part-nsf-proposal-summary-page/

3. Project Description (6-12 pages):

Research Proposal

A. Background and literature review

- What is the background needed to appreciate your proposed research?
- What is the present state of knowledge in your field?
- What is the gap in knowledge?
- How will your proposed research address this gap?

B. Research Strategy

- Most proposals will be separated into several specific aims, each of which addresses a separate hypothesis
- What approach(es), materials, and methods will be used for each aim?
- How will the data be analyzed?
- How will the results be interpreted in light of the hypothesis?
- What are potential pitfalls or alternative approaches that could be used?

C. Significance of the research
- What impact will this proposal have for the field?

References Cited
Reference information is required in the text of the proposal as citations and in the references cited section. Any academic citation format is acceptable, but must include the names of all authors (in the same sequence in which they appear in the publication), the article and journal title, book title, volume number, page numbers, and year of publication.

Optional components (determined by Advisor)
An advisor may require their student to produce other documents (ex: a more detailed methods section, comprehensive background/introduction, timeline, or additional tables/figures).

Dissertation proposal defense

Selection of Committee Chair: The student will select a committee member to head their committee. The Chair cannot be the dissertation advisor and must have a member of the Biological Sciences Department. The Chair's role is to provide guidance for the committee and to be a resource for the student should any issues arise. The Committee Chair must agree to and accept this role.

Timing of the Proposal Defense: The oral proposal defense should occur before the end of the 6th semester and cannot occur in the same semester as the dissertation defense and graduation. The written proposal must be submitted to the committee at least two weeks before the oral defense.

Scheduling and Advertising: It is the responsibility of the student, with assistance from the Biology Administrator and/or mentor, to reserve an appropriate venue which can comfortably accommodate the committee and audience for the defense. The room should be reserved for at least 2 hours. The date, time, and defense title should be shared with the Biology Administrator at least two weeks in advance so that the defense can be advertised within the department and University News (note if this step is not taken, the proposal defense can be canceled by the Graduate Coordinator). If external committee members are attending the defense remotely, it is the student's responsibility to create and share a link for the meeting and to reserve the technology necessary (e.g.: the Owl) for a successful hybrid meeting.

Oral proposal Structure: The proposal defense is open to the public and will be divided into five sections. Interruptions from the audience are discouraged (however, brief critical clarifications/question from the committee are allowed)

1) Dissertation presentation (~45min, open to public)
2) Questions from public (open to public)
3) Questions from committee (open to public)
4) Closed door questions (private: student, PI, committee)
5) Committee convenes (private: PI and committee)
6) Decision delivered to student (private: student, PI, committee)

Outcome: There are three possible outcomes of the exam: Pass, Conditional Pass, and Fail. The chair will deliver the results to the student by collecting statements/suggestions from committee members. The chair will also initiate and sign the **Report of Examination Form**, available via Docusign (see Appendix III). The report must then be signed by the PI/advisor, the committee members, the student, and the Graduate Coordinator after completion of the proposal defense.The committee will set the terms and timeline for the student to convert a Conditional Pass to a Pass. If the student fails, an opportunity for re-examination may be granted. The committee will outline requirements and recommendations for the student to address prior to re-examination. If students are unable to successfully defend their proposal after a second attempt, they will be advised to pursue completion of a Master's degree.

PI's role during the defense: The PI will introduce the student. As the aim of the defense is to examine the student's preparedness and scientific knowledge, the PI should otherwise refrain

from speaking or giving assistance/visual cues to the student during the defense. The Chair will deliver the decision to the student.

Committee Chair's role: At the beginning of the defense, the Committee Chair will remind the committee and audience of the rules for the defense. After the presentation has concluded, the Chair opens public questioning (step 2). After the audience has had time to ask questions, the Chair invites the public to stay or leave prior to committee questions (step 3) and asks the public to leave for the closed door session (step 4). While all committee members are encouraged to avoid sensitive questions during the public sessions, the Chair can defer questions to the closed door session. The chair then leads the closed door session and deliberations. The Chair will deliver the final outcome to the student.

Annual committee meetings

Committee meetings are an essential component of the dissertation process and are an excellent opportunity to get feedback and advice from outside researchers. These meetings are often less formal than presentations/defenses. While the student should prepare slides to present their work/plans and guide the discussion, the meeting is more of a conversation than a presentation.

Timing and scheduling: After the proposal defense, the committee will meet at least once a year, or more often if needed. The committee meeting should be booked well in advance to assure all members can be present. It is the responsibility of the student, with assistance from the Biology Administrator and/or mentor, to reserve an appropriate venue or create and share a link for external members attending virtually. The room should be reserved for 2 hours.

Goals: The goal of committee meetings is for the student to update the committee on progress and discuss any changes to the dissertation plan. In addition, it is an opportunity for the committee to provide insight and guidance to the student. The student should prepare an informal ~45 minute presentation providing updates and questions since the previous year's meeting as well as plans for upcoming work.

Meeting structure: Committee meetings are more of a discussion than a traditional presentation. The student will present their work and future plans to the committee, who may interrupt at any time with questions/comments. The student is not expected or required to provide refreshments for this meeting.

PI's role in the meeting: The PI introduces the student. As the meeting is meant to assess student's performance and progress, the proceedings should be led by the student, with the PI serving in a support capacity to help clarify if needed.

Committee Chair's role in the meeting: The Chair will meet with the student one-on-one in advance of the Committee meeting and will provide opportunities for the student to bring up any issues in their lab or with their PI that need addressing. The Chair will assure that the committee drives the discussion based on the student's presentation and any questions or issues that arise.

Report: A Dissertation Committee Meeting Report (see Appendix II) will be completed by the student and committee upon completion of each meeting, listing updates and action items as well as specific milestones and a detailed time-line for completion. These reports will be approved by the student's committee, submitted to the graduate coordinator and included within the student's academic record.

Written dissertation

The dissertation must satisfy the following criteria: 1) it should demonstrate the candidate's intellectual competence and maturity in the field of concentration; 2) it should make an original and valid contribution to knowledge; and 3) it should be an individual achievement and the product of independent research. Although doctoral dissertations may result from a project involving collaboration of several scholars, the individual contribution of each doctoral candidate must be substantial, clearly identifiable, and presented separately. The committee will judge the completed dissertation in terms of the candidate's ability to review and make critical use of the literature; to formulate a problem, develop appropriate methodology, and work systematically toward a solution; and to summarize the material or data and draw conclusions from them. The writing should be of publishable quality.

Formatting:
Dissertations are filed with the University and must meet current formatting guidelines, which can be found at https://www.uml.edu/catalog/graduate/policies/dissertation-thesis/.
1. Standard 8.5" x 11" page size
2. 12-point, Times New Roman font
3. 10-point font may be used for references, footnotes, figure captions and text within figures
4. 1" margins on all sides
5. Double-spaced (approximately 5 lines per inch) or greater line spacing. Do not use line spacing options such as "exactly 12 point," that are less than single spaced.

Structure: The multi-chapter dissertation has at its core a set of chapters (at least 3, but may be up to six) each judged by the dissertation committee to be of a quality and form suitable for publication in the peer reviewed literature of a relevant field.

1. **Title page and Abstract**
2. **Dedication, Acknowledgements**
3. **Table of Contents**
4. **Introduction**: The purpose of the introductory chapter is to explain the background or context in which the research is set, and to lay out its significance for the field. This chapter serves as an overview to connect the individual monographs together. It provides an argument justifying the choice of dissertation topic and would typically include a literature review demonstrating that the research was needed, not yet performed by others, and has a reasonable chance of achieving its stated objectives.
5. **Body** - The chapters, which form the body of the dissertation, are of publishable quality but they are often longer than the currently acceptable length of papers in most journals. This permits the dissertation to serve its archival value, documenting the detailed research methods and results more fully than journal publication will permit. The level of detail in the chapters will be agreed upon by the candidate and the dissertation advisor and/or committee. Each chapter must contain the following:
    a. Introduction
    b. Methodology
    c. Results
    d. Discussions

    e. Conclusions
    f. Literature Cited
6. **Conclusion**: The dissertation's concluding chapter provides the opportunity to present the broad conclusions of the body of work because it focuses on implications for the field as a whole, rather than simply on the next steps in the research process. In addition, this chapter presents the necessary opportunity to discuss the interdependence of the individual chapters.
7. **Appendix**
8. **Biographical Sketch** of Author (required)

Dissertation preparation and submission: For more details about these official UML guidelines, see here:
https://www.uml.edu/catalog/graduate/policies/dissertation-thesis/format.aspx
Every graduate student who completes a dissertation is required to bear the cost of the archival fee. Copywriting is optional and available for an additional fee.

Dissertation defense

Scheduling and Advertising: It is the responsibility of the student, with assistance from the Biology Administrator and/or mentor, to reserve an appropriate venue for the defense. The room should be reserved for at least 2 hours. At least 2 weeks prior to the dissertation defense, announcements of the defense that list the graduate student's name, dissertation title, and place and time of the defense, must be submitted to the following: UMass Lowell Today, Chair of the department, College Dean, Graduate School Dean for posting and distribution. (Note if this step is not taken, the proposal defense can be canceled by the Graduate Coordinator).
If external committee members are attending the defense remotely, it is the student's responsibility to create and share a link for the meeting and to reserve the technology necessary (e.g., the Owl) for a successful hybrid meeting.

Oral Defense Structure: The defense is open to the public and will be divided into five sections. Interruptions from the audience are discouraged (however, brief critical clarifications/question from the committee are allowed)

1) Dissertation presentation (45-60min, open to public)
2) Questions from public (open to public)
3) Questions from committee (open to public)
4) Closed door questions (private: student, PI, committee)
5) Committee convenes (private: PI and committee)
6) Decision delivered to student (private: student, PI, committee)

Outcome: There are three possible outcomes of the exam: Pass, Conditional Pass, and Fail. The chair will deliver the results to the student by collecting statements/suggestions from committee members. The chair will also initiate and sign the **Report of Examination Form**, available via Docusign (see Appendix III). The report must then be signed by the PI/advisor, the committee members, the student, and the Graduate Coordinator after completion of the dissertation defense. The committee will set the terms and timeline for the student to convert a Conditional Pass to a Pass. If the student fails, an opportunity for re-examination may be granted. The committee will outline requirements and recommendations for the student to address prior to re-examination.

PI's role during the defense: The PI will introduce the student. As the aim of the defense is to examine the student's preparedness and scientific knowledge, the PI should otherwise refrain from speaking or giving assistance/visual cues to the student during the defense. The chair will deliver the decision to the student.

Committee Chair's role: At the beginning of the defense, the Committee Chair will remind the committee and audience of the rules for the defense. After the presentation has concluded, the Chair opens public questioning (step 2). After the audience has had time to ask questions, the Chair invites the public to stay or leave prior to committee questions (step 3) and asks the public to leave for the closed door session (step 4). While all committee members are encouraged to avoid sensitive questions during the public sessions, the Chair can defer questions to the closed door session.

Applying for graduation

Applying for graduation is a multi-step process. At the beginning of the semester a student intends to graduate, they must first submit a "Declaration of Intent to Graduate Form" which is available at: https://www.uml.edu/thesolutioncenter/Forms/Academic-Forms.aspx to the Biological Sciences Graduate Coordinator.

The Graduate Coordinator checks the student's records. If all is in order, the student's application for graduation will be submitted to the university. The Intent to Graduate Form must be received by a specific date for Spring Graduation or Fall Graduation. If additional action is required (i.e., there is a problem with the number of credits towards the degree or the student's GPA), the student receives a letter specifying what actions are required to graduate.

Appendix I. Thesis/Dissertation Enrollment form

A PDF of this form will be provided to you to enroll in dissertation research.

University of Massachusetts Lowell
UMASS Lowell

Department of Biological Sciences

Thesis/Dissertation Enrollment Form

Dissertation research is a requirement for the Applied Biology PhD program. During thesis (BIOL.7430) or dissertation (BIOL.7530) work, the student concentrates on an in-depth, independent, scholarly investigation of a contemporary biological problem. It is the student's responsibility to ensure that this contract is filled in and approved before the project is commenced.

Student Name _____ ID# _____

The student must consult with the research advisor and the graduate student manual to assure proper committee make up.

Committee Members

Biological Sciences Faculty Advisor _____

Committee Chair _____

Second Committee member _____

Additional Committee member (if needed) _____

Additional Committee member (if needed) _____

The student presents to the Committee a proposal of intended research and obtains the Committee's approval of the research topic. After completing the written thesis, the student gives an oral presentation of her/his results to the Thesis Committee.

Projected Proposal Presentation Date _____

Faculty Advisor _____ _____
Print Signature

Graduate Student _____ _____
Print Signature

By signing above the student confirms that permission has been obtained from his/her thesis advisor to complete the MS Biological Sciences thesis requirements. The student also agrees to comply with posted guidelines for thesis research and UML policies concerning academic integrity.
https://www.uml.edu/Catalog/Graduate/Policies/Grading-Policies.aspx
https://www.uml.edu/Catalog/Graduate/Policies/Academic-Integrity.aspx

Graduate Coordinator _____ _____
Print Signature

*The Graduate Coordinator (or Department Chair if necessary) signature affirms approval of the Department Graduate Academic Committee.

Appendix II. Dissertation Committee Meeting Report

A PDF of this form will be provided to you for Dissertation Committee Meetings.

University of Massachusetts Lowell

UMASS

Department of Biological Sciences

Dissertation Committee Meeting Report

The student should provide this form to the committee for the meeting. It is to be completed and signed by all upon conclusion of the meeting. The student shall submit a copy of the completed form to the Biological Sciences Graduate Coordinator and committee members. It is the student's responsibility to ensure that a report is submitted to the committee and feedback provided by the committee ***at least once per year***.

Student Name _____ ID# _____

Date of report _____

Last report submitted _____

Next meeting date _____

Committee Member Certification

	Print	Signature
Biological Sciences Faculty Advisor		
Departmental Committee member		
Second Committee member		
Additional Committee member (If needed)		
Additional Committee member (If needed)		

By signing above the committee confirms that the student has presented a progress report for committee review.

Summary of progress since last report (attach separate pages if necessary).

Committee recommendations (attach separate pages if necessary).

	Print	Signature
Graduate Student		
Graduate Coordinator		

Appendix III. Report of Examination

Your committee chair will initiate the Docusign version of this form (linked above) upon completion of your qualifying exam, proposal defense, or dissertation defense.

University of Massachusetts Lowell
UMASS

Department of Biological Sciences

Report of Examination

This form should be used for MS Thesis Proposal Presentation/Defense, Ph.D. Proposal/Dissertation Defense/Oral Exam. The student must present this form to the committee on the date of the presentation/exam. PASS should only be recorded in the case where ALL portions of the exam are completed. In the case of FAIL, please indicate recommendations concerning the earliest opportunity for re-examination, if this possibility is to be granted.

Student Name _____ ID#_____

Examination/Presentation Date _____

Indicate PASS or FAIL in the appropriate box.

MS Proposal Presentation	☐ PASS	☐ FAIL	☐ Conditional PASS
MS Thesis Defense	☐ PASS	☐ FAIL	☐ Conditional PASS
Ph.D. Proposal Defense	☐ PASS	☐ FAIL	☐ Conditional PASS
Ph.D. Dissertation Defense	☐ PASS	☐ FAIL	☐ Conditional PASS
Qualifying Exam	☐ PASS	☐ FAIL	☐ Conditional PASS

Recommendations:

Committee Members Present

Biological Sciences Faculty Advisor _____ (Print) _____ (Signature)

Departmental Committee member _____ (Print) _____ (Signature)

Second Committee member _____ (Print) _____ (Signature)

Additional Committee member (If needed) _____ (Print) _____ (Signature)

Additional Committee member (If needed) _____ (Print) _____ (Signature)

By signing above the committee certifies the exam results.

Student Name _____ (Print) _____ (Signature)

Graduate Coordinator _____ (Print) _____ (Signature)

Appendix IV. Qualifying Exam Rubric

Applied Biology PhD Qualifying Exam Rubric

Ratings for each category will be given as an overall impression for all three papers.
To pass the qualifying exam, each category must receive a rating or Satisfactory or higher.

	No proficiency demonstrated **Failure (1)**	Marginal proficiency demonstrated **Unsatisfactory (2)**	Proficiency demonstrated **Satisfactory (3)**	Exceptional proficiency demonstrated **Excellent (4)**	**Rating (1-4)**
Understanding of hypothesis & significance	Failed to articulate the significance of the work and identify testable hypotheses	Superficial demonstration of the significance of the work by providing a basic argument that defends stated hypotheses.	Satisfactory demonstration of the significance of the work by providing a compelling argument that defends the stated hypotheses.	Exceptional demonstration of the significance of the work, including effectively providing defense and abstract implications of the state hypotheses.	
Experimental strategies & methods	Failed to present reasoning for the strategy and why methods were appropriate for the scope of the study.	Undeveloped understanding of why strategy and methods were appropriate for the scope of the study.	Satisfactory understanding of strategy and methods, including basic explanations for why specific methods were chosen, and alternative approaches available.	Fully developed understanding of strategy and methods, including advanced explanations for why specific methods were chosen, detailed explanations of the principles on which methods work, and why alternative approaches were not chosen.	
Data analysis and interpretation	Failed to explain how data would be collected and analyzed so as to derive valid conclusions and possible experimental outcomes.	Superficial explanation of how data collection and analysis led to conclusions; no possible experimental outcomes proposed.	Satisfactory explanation of data collection and analysis, including possible experimental outcomes and statistical analyses that demonstrate validity of conclusions.	Exceptional explanation of data collection and analysis, including possible outcomes and insights impacting the general field of study, and statistical analysis demonstrating validity of conclusions.	
Verbal communication	Failed to communicate ideas or support conclusions. Answers lacked logical flow, reasoning, and support of arguments	Marginally effective communication of findings and results, often lacking clarity due to missing details. Answers have some logical flow and include basic reasoning and support of arguments.	Satisfactory communication, with clarity and expected detail, of findings & results. Answers include logical flow, including an articulation of reasoning and thorough support of arguments.	Highly effective communication, including general implications of results in relation to the field of study. Answers provided excellent logical flow, reasoning, and support of arguments, including solid interpretations using a clear train of logic.	
Knowledge directly related to topic	Displays major gaps in the knowledge directly related and specific to papers and/or demonstrated major conceptual failings.	Displays superficial knowledge related to papers, including minor gaps in knowledge and/or minor conceptual failings.	Displays adequate knowledge of the field and showed no conceptual failings. Demonstrated a grasp of the problem and the papers assigned.	Displays comprehensive knowledge of the field and demonstrated the ability to integrate relevant concepts to produce creative insights, including a grasp of relevant literature beyond the paper assigned.	
Knowledge indirectly related to topic	Failed to demonstrate general knowledge expected of students at this point in their training.	Superficial general knowledge expected of students at this point in their training.	Satisfactory general knowledge expected of students at this point in their training.	Exceptionally comprehensive general knowledge that exceeds typical students at this point in their training.	
Overall critical thinking	Failed to demonstrate any ability to approach scientific questions with rational experimental strategies.	Superficial ability to approach scientific questions with rational experimental strategies.	Satisfactory ability to approach scientific questions with rational experimental strategies.	Exceptional ability to approach scientific questions with rational experimental strategies and insights that exceed students at the current stage of training.	
The overall rating reflects the student's total performance for all three papers. It should be consistent with, but is not a mathematical average of, the individual ratings shown above that may each carry different weight.				**overall rating**	

Appendix V. Example Performance Improvement Plan (PIP)

Student: xxx
PI: xxx

Reason for pursuing an improvement plan: xxx

PIP effective date: xxx
PIP end date: xxx

PIP activities – these should be specific and include concrete deliverables.

1. Example: weekly – due by 5p the day before individual meeting: agenda with discussion points and data summary. Should include the following: what I did this week; data summary; points I need help with; what I plan to do next week.
2. Example: weekly – at individual meetings, a breakdown of activities per hour, categorized into: TA duties, classes, research (further broken down into weekly meetings, bench work, reading, writing).
3. Example: monthly – at the end of each month, a self-reflection detailing what went well, where you struggled, and how you'll address these issues in the future. This will be discussed at our weekly meeting
4. Example: by Jun 1 – troubleshooting protein extract prep for anti-H3 western followed by three replicates comparing double mutants to controls.

Outcome assessments:

1. Example: Minimum of B in all Fall courses.
2. Example: Satisfactory evaluation as TA from instructor of record.
3. Example: Demonstrated improvement in research productivity:
    a. Outline of dissertation proposal
    b. Annotated bibliography for proposal introduction
    c. Generation of Fig 1 (anti-H3 western) and Fig 2 (germline immunofluorescence)

Note that failure to meet these outcomes may result in termination of the graduate position. If only some, but not all, of the outcomes are met, your PI may consider transfer to the MS program (with or without funding) by XXX. The choice to continue progress as a doctoral student, transfer to the MS program, or terminate position as a graduate student is at the discretion of the PI and the Graduate Coordinator, and will be based on the outcome assessment listed above.

	Signature	Date
Student		
PI		
Graduate coordinator		

Appendix VI. Important milestones for the Applied Biology Ph.D.

	Fall	Spring	Summer
Year 1	☐ Courses (full time is 9 credits). ● Suggested: ○ BIOL.6060 App Bio I (1 cr) ○ BIOL.6030 Colloq (1 cr) ○ Electives (7 cr) ☐ If necessary, enroll in dissertation credit (not encouraged).	☐ Courses (full time is 9 credits). ● Suggested: ○ BIOL.6070 App Bio II (1 cr) ○ BIOL.6030 Colloq (1 cr) ○ Electives (7 cr) ☐ If necessary, enroll in dissertation credit (not encouraged).	☐ Conduct research full-time. ● discuss summer salary with PI.
Year 2	☐ Courses (full time is 9 credits). ● Suggested: ○ BIOL.6040 Prof Comm (3 cr) ○ BIOL.7210 Special Prob (1-3 cr) ○ Electives (3-5 cr) ☐ Select Dissertation Advisory Committee, in discussion with PI. ● Submit Dissertation Enrollment Form (App I). ☐ If necessary, enroll in dissertation credit (not encouraged).	☐ Courses (full time is 9 credits). ● Suggested: ○ BIOL.6050 Prop Writing (1 cr) ○ BIOL.7210 Special Prob (1-3 cr) ○ Electives (5-7 cr) ☐ **Qualifying exam** (Rubric in App IV). ● March – choose committee ● May-June – qualifying exams ● Submit Report of Examination (App III). ☐ Write dissertation proposal with PI. ☐ If necessary, enroll in dissertation credit (not encouraged).	☐ Option to **defend dissertation proposal.** ● Proposal defense must be open to public; at least two weeks before, schedule & advertise via UML. ● Submit Report of Examination (App III). ☐ Conduct research full-time.
Year 3+	☐ Option to defend dissertation proposal, if not defended in year 2. ☐ Conduct research (enroll in dissertation credit).	☐ Schedule annual **committee meeting.** ● Submit Dissertation Committee Meeting Report (App II). ☐ Conduct research (enroll in dissertation credit).	☐ Conduct research full-time.
Final year	☐ In penultimate semester, write dissertation with PI (UML guidelines). ☐ Conduct research (enroll in dissertation credit).	☐ In final semester, **defend** dissertation. ● At start of semester, must submit Declaration of Intent to Graduate form to determine eligibility. ● Defense must be public; at least two weeks before defense, schedule & advertise via UML. ● Submit Report of Examination (App III). ☐ Conduct research (enroll in dissertation credit).	☐ Live your life as a newly minted Ph.D.!